

Multicollinearity in Multivariable Linear Regression

1 Introduction

Let's solve a **multivariable linear regression** problem step by step with 3 data points. We'll have two independent variables X_1 and X_2 , and one dependent variable Y .

2 Given Data

X_1	X_2	Y
1	2	5
2	3	7
3	4	9

We want to model the relationship:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

3 Step 1: Calculate the Means

The mean values for each variable:

$$\bar{X}_1 = \frac{1+2+3}{3} = 2, \quad \bar{X}_2 = \frac{2+3+4}{3} = 3, \quad \bar{Y} = \frac{5+7+9}{3} = 7$$

4 Step 2: Compute Covariances and Variances

We compute the following:

- Covariance $\text{cov}(X_1, Y)$
- Covariance $\text{cov}(X_2, Y)$
- Covariance $\text{cov}(X_1, X_2)$
- Variance $\text{var}(X_1)$
- Variance $\text{var}(X_2)$

4.1 Covariance Calculation

The covariance formula:

$$\text{cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

For X_1 and Y , we compute:

$$(X_1 - \bar{X}_1) = [1 - 2, 2 - 2, 3 - 2] = [-1, 0, 1]$$

$$(Y - \bar{Y}) = [5 - 7, 7 - 7, 9 - 7] = [-2, 0, 2]$$

$$(X_1 - \bar{X}_1)(Y - \bar{Y}) = [-1 \times -2, 0 \times 0, 1 \times 2] = [2, 0, 2]$$

Summing up:

$$2 + 0 + 2 = 4$$

Dividing by $n - 1$:

$$\text{cov}(X_1, Y) = \frac{4}{2} = 2$$

Similarly, for X_2 and Y , we find:

$$\text{cov}(X_2, Y) = 2$$

For X_1 and X_2 , we find:

$$\text{cov}(X_1, X_2) = 1$$

4.2 Variance Calculation

The variance formula:

$$\text{var}(X) = \frac{\sum (X_i - \bar{X})^2}{n - 1}$$

For X_1 :

$$\text{var}(X_1) = \frac{2}{2} = 1$$

For X_2 :

$$\text{var}(X_2) = \frac{2}{2} = 1$$

5 Step 3: Compute Coefficients β_1 and β_2

$$\beta_1 = \frac{\text{cov}(X_1, Y)\text{var}(X_2) - \text{cov}(X_2, Y)\text{cov}(X_1, X_2)}{\text{var}(X_1)\text{var}(X_2) - \text{cov}^2(X_1, X_2)}$$

Substituting values:

$$\beta_1 = \frac{2 \times 1 - 2 \times 1}{1 \times 1 - 1^2} = \frac{2 - 2}{1 - 1} = \frac{0}{0}$$

Since the denominator is zero, the calculation fails due to ****multicollinearity****, meaning X_1 and X_2 are highly correlated.

6 Conclusion

Multicollinearity prevents us from computing β_1 and β_2 properly. In practice, alternative methods such as **Principal Component Regression (PCR)** or **Ridge Regression** are used to handle multicollinearity in regression models.